team_SGS · Paytm Hackathon · AI-Powered Financial Journeys

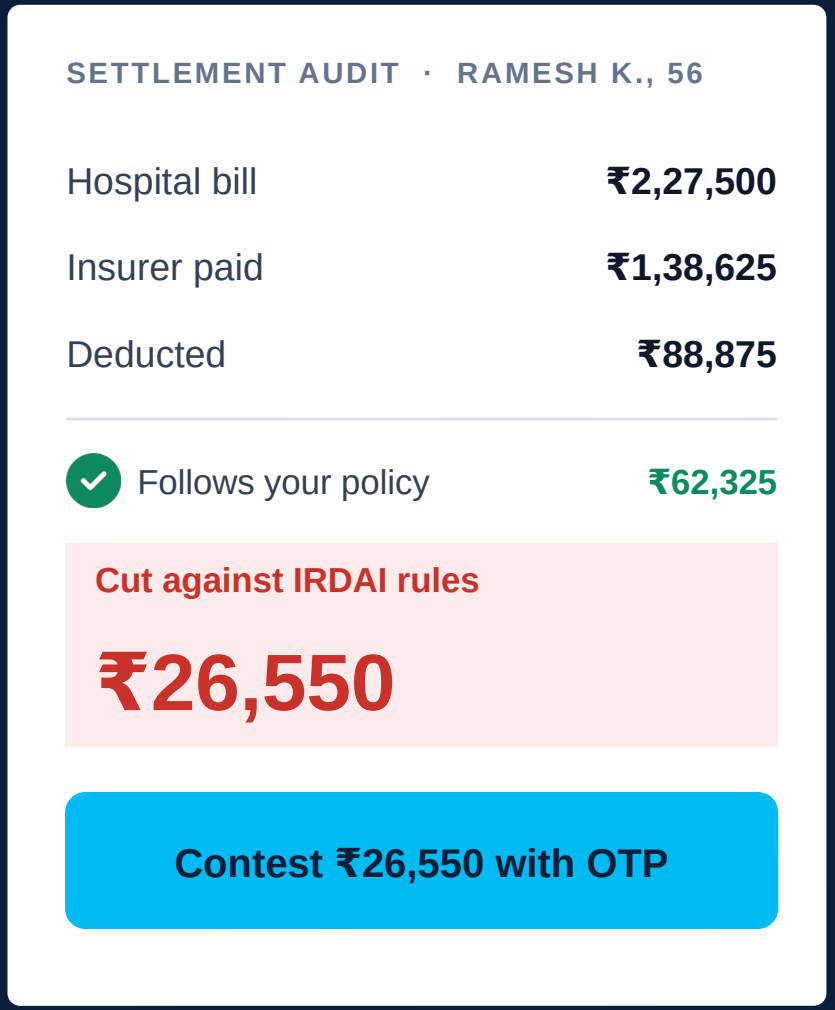
Haq

हक़ · what is rightfully yours

An AI claim agent on the patient's side of a health insurance claim

Reads your policy and bills, checks each deduction against IRDAI rules, files with your OTP and hands judgement calls to a human expert

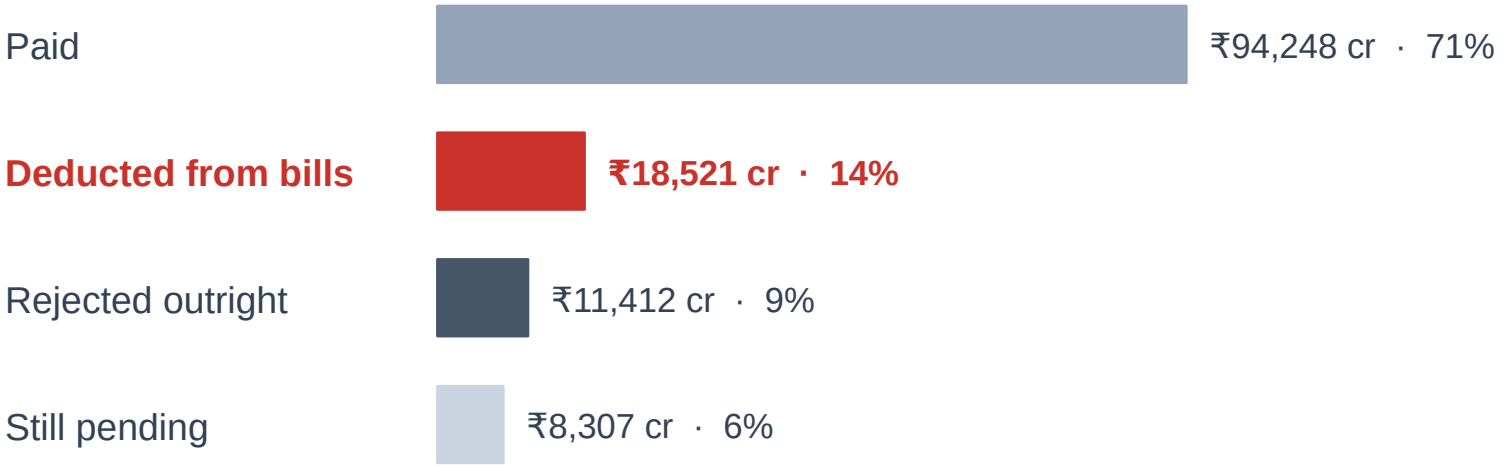
Tarun Bhutra · Aayush Jha



THE PROBLEM

Insurers cut ₹18,521 crore from hospital bills in FY25, and patients rarely learn why

Where ₹1,32,488 crore of health claims went in FY25



Deductions rose 23% from ₹15,100 cr in FY24 and now exceed outright rejections. IRDAI records no reason for any cut, so the family cannot tell a legal deduction from an illegal one.

>50%
of claimants saw a rejection or a partial payout
LocalCircles, Jan 2025, 28,700 claimants

94.8%
of valid health complaints to the Ombudsman are about claims cut or refused
Insurance Ombudsman Annual Report 2023-24

47%
say they personally faced deliberate delays from insurers
LocalCircles, Jan 2025

Source: IRDAI Annual Report 2024-25, Table I.29; Business Standard (FY24); Insurance Business Asia, 27 Jul 2026; LocalCircles; CIO Annual Report 2023-24

WHY THE JOURNEY FAILS

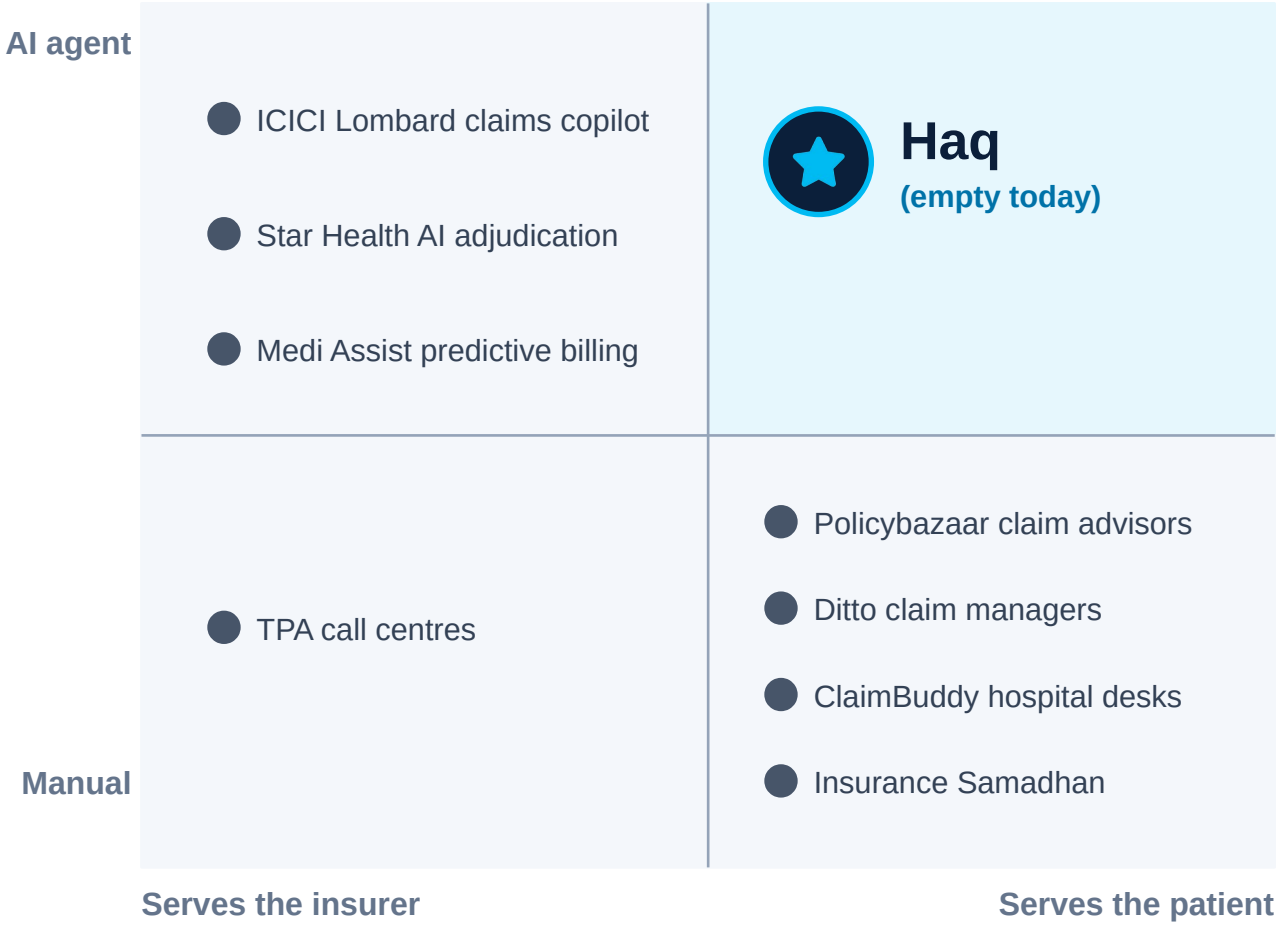
A family decides blind at six points between the admission desk and the Ombudsman

1 Admission desk	2 Pre-authorisation	3 Final bill	4 Discharge	5 Settlement	6 Grievance
Picks a room in minutes, under stress	Waits while hospital and TPA exchange forms	Hospital assembles the bill and discharge file	Doctor says go home; the family waits	Letter lists deduction codes, not reasons	Complains to whoever answers first
65%	**13%**	**3–4 h**	**6–48 h**	**83%**	**21%**
don't understand their own policy	of cashless approvals take more than 1 hour	typical billing time before the insurer's clock even starts	wait for 50–60% of patients, despite a 3-hour rule	want a transparent way to track their claim	of Ombudsman health complaints rejected for the wrong order
MISSING	MISSING	MISSING	MISSING	MISSING	MISSING
The room-rent cap and what it does to the whole bill	**Live status and the 1-hour clock**	**Who is holding the file, right now**	**That the insurer pays for delay beyond 3 hours**	**Which cuts follow the policy and which break rules**	**The right order: insurer, Bima Bharosa, Ombudsman**

Source: CoverSure 2025; Lok Sabha reply, 1 Dec 2025; Medi Assist CEO (The Federal, 2024); LocalCircles 2025–26; CIO Annual Report 2023-24

THE GAP

Claims AI in India works for insurers; the patient's side still runs on phone calls



Rights patients already have, unenforced on their behalf

Cashless decision within 1 hour
IRDAI Health Master Circular, 2024

Discharge approved within 3 hours, or the insurer pays for the extra stay
IRDAI Health Master Circular, 2024

The hospital sends the documents, not the patient
IRDAI Health Master Circular, 2024

No proportionate cut on medicines, consumables, implants, diagnostics or ICU
IRDAI/HLT/REG/CIR/151/06/2020

No non-disclosure challenge after 60 months of continuous cover
Moratorium clause, all health policies

Source: Microsoft (ICICI Lombard); Angel One (Star Health, Mar 2026); The Federal (Medi Assist); company sites; IRDAI Master Circular 29 May 2024; IRDAI circular 151/06/2020

OUR SOLUTION

Haq follows one hospital stay from the admission desk to the last rupee



BEFORE ADMISSION

Know the cost first

Reads your policy, prices each room on offer and shows your out-of-pocket cost before you choose

Policy reading · simulation



DURING THE STAY

Keep IRDAI's clocks

Starts the 1-hour and 3-hour timers, nudges the hospital desk and records any breach as evidence

Tracking · alerts



AFTER SETTLEMENT

Audit every rupee

Reads the bill and settlement letter, recomputes your entitlement and flags cuts that break the rules

Document AI · rules engine



IF MONEY IS WITHHELD

Act and escalate

Drafts the contest, sends it on your OTP, then runs insurer → Bima Bharosa → Ombudsman in order

Tool calls · scheduling



Human claims expert, one tap away on every screen. They receive the full case file: documents, calculations, drafts and clock status, so the family does not repeat the story.

Channels: Paytm app, WhatsApp and voice, in Hindi and regional languages

BEFORE VS AFTER

Same hospital stay, two outcomes: with Haq, Priya contests ₹26,550 and sees each cost first

Demo case: Priya's father Ramesh, 56, gallbladder surgery, 4 days, ₹5 lakh policy since 2019

	Today	With Haq
Admission	Picks the ₹8,000 room without knowing the ₹5,000 cap	Sees ≈ ₹62,300 out of pocket vs ≈ ₹5,700 for twin-share, then chooses
Discharge	Waits into the evening and pays for the extra hours	3-hour clock runs; breach logged, so the insurer bears the extra charges
Settlement	"₹88,875 deducted" with no reasons	₹62,325 explained line by line; ₹26,550 flagged against IRDAI rules
Response	Gives up, or calls the helpline for weeks	Contest drafted with citations and sent on OTP in under 10 minutes
Escalation	Complains to the wrong forum first	Ladder scheduled: insurer, day 14 Bima Bharosa, then Ombudsman
Outcome	**₹88,875 absorbed · no contest · no clock**	**₹26,550 contested · filed in < 10 min**

Source: Synthetic demo case; entitlement rules from IRDAI Master Circular 2024 and circular 151/06/2020

HOW THE AI WORKS

An orchestrator plans, specialists read, a rules engine does the maths and a verifier checks every word

Channels: Paytm app · WhatsApp · Voice (AI Soundbox / IVR)

Case Orchestrator

One stateful case per hospital stay: intake → cover → admit → discharge → settle → contest → escalate

Intake & Consent
Identity, DPDP consent, fetches policy

Policy Intelligence
Wording → coverage graph with clause IDs

Document Intelligence
Vision OCR of bills and letters, line mapping

Entitlement Engine
Deterministic code, no LLM: every rupee

Clock Keeper
IRDAI timers, breach detection

Advocate
Drafts, files after OTP, schedules ladder

Verifier → Communicator: amounts must match the engine · every claim cites a clause · AI disclosed · PII redacted · plain Hindi or English

Memory & knowledge: case store and append-only audit log · clause and circular index (pgvector) · versioned IRDAI rulebook · encrypted document vault

Tools and APIs

- `NHCX FHIR claim API`
- `Paytm policy record`
- `Hospital tariff`
- `DigiLocker / ABHA`
- `Email · insurer portal`
- `Bima Bharosa · Ombudsman`
- `Scheduler · notifications`

Human expert console

Takes over with the full case file. Triggers: medical necessity, fraud, disputes above ₹1 lakh, low confidence

Models: Claude Sonnet 5 plans and drafts; Claude Haiku 4.5 classifies and verifies; Claude vision reads bills

ACTION LAYER

Haq reads, computes, files, tracks and escalates through audited tool calls

Read

- `fetch_policy()`
- `parse_coverage()`
- `ocr_document()`

Room cap ₹5,000/day, cited to clause D.1(i)

Compute

- `simulate_rooms()`
- `engine.recompute()`
- `verifier.check()`

Entitled ₹1,65,175 against ₹1,38,625 paid

File

- `draft_contest()`
- `request_otp()`
- `send_grievance()`

Contest sent to the insurer's grievance officer

Track

- `clock.start()`
- `nhcx.poll_status()`
- `detect_breach()`

Discharge clock breached at 3 h 01 m, evidence saved

Escalate

- `schedule_ladder()`
- `file_ombudsman()`
- `handoff_expert()`

Bima Bharosa draft queued for day 14

AGENT TRACE FROM THE DEMO (mock insurer API, synthetic bill)

```
10:42:03   ocr_document(settlement_letter, bill)          → 9 line items, confidence 0.97
10:42:05   engine.recompute(rules=R1..R6)                 → 3 violations, recoverable ₹26,550
10:42:06   verifier.check(amounts, citations)             → pass: numbers match engine, 3 clauses cited
10:43:10   send_grievance(to=insurer_GRO)  [OTP approved] → filed, ref GRV-2291, ladder scheduled D+14
```

Source: Haq demo build; tool names as implemented in the orchestrator

TRUST, SAFETY & HUMAN ESCALATION

Haq files only with your approval, the insurer still decides, and judgement calls go to a human

Haq does on its own

- Reads policies, bills and letters
- Computes what you are owed and what you will pay
- Runs clocks, reminders and status checks
- Explains every result in your language

Only with your OTP approval

- Sends a contest or grievance
- Shares documents with the insurer
- Files with Bima Bharosa or the Ombudsman
- Edits stay open until you approve

Always goes to a human expert

- Disputes over medical necessity
- Allegations of non-disclosure or fraud
- Full rejections, or disputes above ₹1 lakh
- Low confidence after one clarification, or when you ask

Guardrails on every message

Says it is an AI

RBI FREE-AI, 2025

Cites a clause for each claim

Moffatt v. Air Canada lesson

Amounts only from the rules engine

LLM never generates ₹

Consent per use, PII redacted

DPDP Act 2023

Every step in an audit log

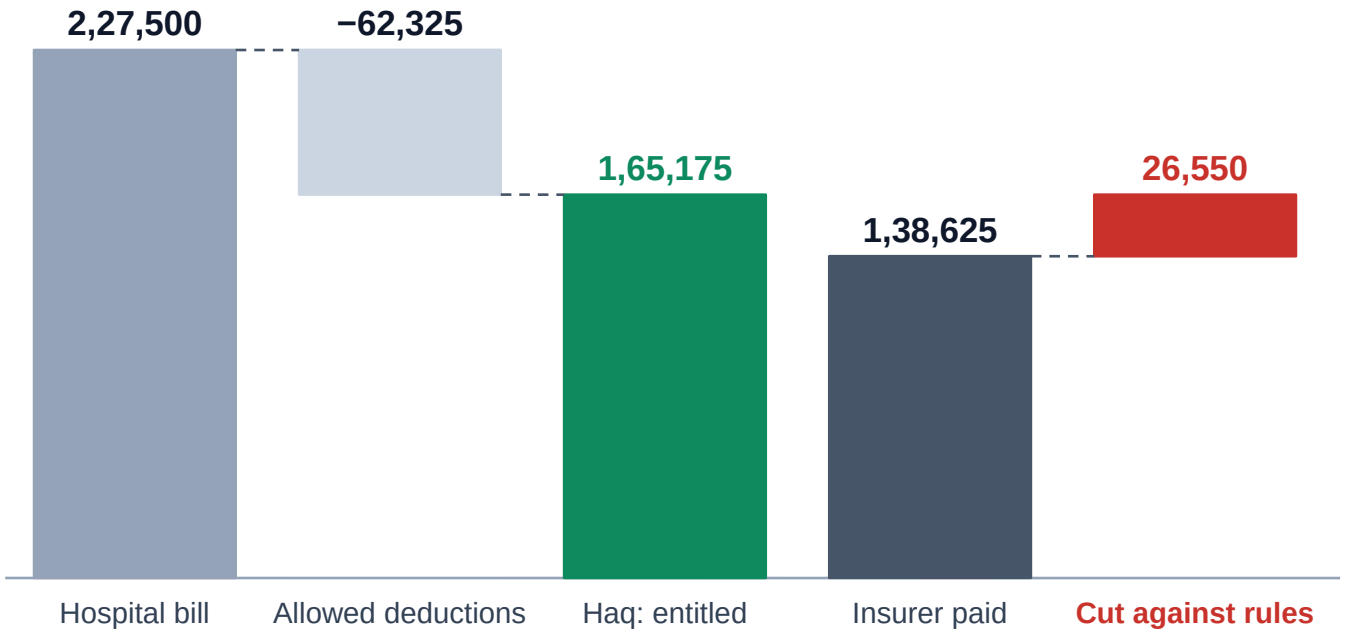
Exportable to insurer or IRDAI

Source: RBI FREE-AI report, 13 Aug 2025; Moffatt v. Air Canada (BC CRT, Feb 2024); DPDP Act 2023; notes cite Klarna (2025) and Gartner (Jul 2024)

DEMO

₹26,550 of the ₹88,875 deducted broke IRDAI rules; Haq filed the contest in under 10 minutes

Ramesh's claim, recomputed by Haq (₹)



Three cuts the rules forbid

The insurer applied the room-rent proportion (0.625) to lines that IRDAI exempts from it

Pharmacy and medicines	₹14,250
Surgical consumables	₹4,050
Diagnostics	₹8,250

Rule: IRDAI/HLT/REG/CIR/151/06/2020. Proportionate deduction applies only to associated medical expenses, never to pharmacy, consumables, implants, diagnostics or ICU.

LIVE DEMO (5 MIN)

1 Cover card | 2 Room sim | 3 Pre-auth clock | 4 Discharge clock | **5 Bill audit** | 6 Contest on OTP | 7 Payout verified | 8 Human handoff

Source: Synthetic bill; policy modelled on IRDAI standard Arogya Sanjeevani wording (UIN UNIHLIP20171V011920); IRDAI circular 151/06/2020

IMPACT

Recovered claims build renewal trust for Paytm, and the same engine sells to insurers

Metric	Pilot target	Counter-metric
Out-of-pocket known before admission	**70% of assisted admissions**	Estimate error under 10%
Rule-breaking cuts contested	**90% within 7 days**	At least 85% of contests upheld
Cases resolved without a human	**60%**	Human reached in under 5 min
Ombudsman complaints rejected for wrong order	**21% → under 3%**	Resolution time not worse

Every metric ships with a counter-metric, so the agent can't win by flooding insurers or blocking humans.

₹185 crore

returned to patients for every 1% of FY25 health-claim deductions that Haq helps recover

For Paytm, as the broker

Claims decide renewals. A broker that recovers money for its customers gives them a reason to renew, and cuts support contacts per claim.

For insurers, via Paytm Intelligence

Run the same audit before settlement to catch rule-breaking cuts before they become Ombudsman cases. Star Health alone drew about 12,000 Ombudsman complaints in FY25.

Source: IRDAI AR 2024-25 (₹18,521 cr deducted); CIO AR 2023-24 (21% non-entertainable); Insurance Business Asia, Feb 2026; targets are team_SGS pilot goals

TECHNOLOGY & ROADMAP

One engine, new rulebooks: credit reports, UPI refunds and motor claims come next

Hackathon stack

Frontend	Next.js mobile web, expert console, trace panel
Agents	FastAPI + Claude Agent SDK, typed tools
Models	Claude Sonnet 5, Haiku 4.5, Claude vision OCR
Data	Postgres + pgvector, append-only audit log
Rails	NHCX FHIR (mock), DigiLocker / ABHA, email
Clocks	APScheduler now, Temporal in production

NOW

Hackathon

Health claims, one policy wording, mock insurer API

→

NEXT 3 MONTHS

Pilot

Paytm health policyholders, one insurer on NHCX, Hindi + 2 languages

→

LATER

Scale

Credit-report disputes, UPI refunds, motor claims; Paytm Intelligence for insurers

Clock and compensation per journey

Journey	Clock	Customer is owed
Health claim (built)	1 h · 3 h · 15 days	Extra stay costs, penal interest
Credit-report error	30 days (21 + 9)	₹100 per day of delay
Failed UPI payment	T+1	₹100 per day of delay
Motor claim	24 h · 15 d · 7 d	Survey and decision on time

Source: RBI credit-information compensation framework (2023); RBI TAT circular (2019); IRDAI motor timelines (Jun 2024); NRCeS NHCX FHIR profiles

APPENDIX

Sources behind the numbers in this deck

- IRDAI Annual Report 2024-25, Table I.29 (health claims FY25)
- Council for Insurance Ombudsmen, Annual Report 2023-24
- IRDAI Master Circular on Health Insurance Business, 29 May 2024
- IRDAI/HLT/REG/CIR/151/06/2020, proportionate deductions
- Arogya Sanjeevani policy wording, UIN UNIHLIP20171V011920
- Lok Sabha reply on cashless authorisation, 1 Dec 2025 (via Upstox)
- LocalCircles health insurance surveys, Jan 2025 and Mar 2026
- CoverSure policy-literacy survey, via Business Today, Jun 2025
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- The Federal, Medi Assist on discharge billing, Nov 2024
- Microsoft AI First Movers: ICICI Lombard claims copilot
- Angel One, Star Health AI claims target, Mar 2026
- Paytm Q1 FY27 earnings release, 20 Jul 2026
- Medianama, Paytm Intelligence, Sep 2026
- RBI FREE-AI Committee report, 13 Aug 2025
- Moffatt v. Air Canada, BC Civil Resolution Tribunal, Feb 2024
- Gartner customer-service AI survey, Jul 2024
- Moneylife, RBI penalises credit bureaus, Sep 2026
- RBI TAT harmonisation circular, 20 Sep 2019

Full list of about 230 sources, with URLs and dates, handed in with the deck